

Parallel Computing: Who Steals Your Speedup

Lab course • Lecture 0

Where we are going today

Theory promises an N -fold speedup. Four thieves take it back: the algorithm, the hardware, the OS, and your own locks.



Every act names one thief, shows you exactly what it steals — and hands you the counter-move. We finish by putting all four on one checklist.

Concurrency is not parallelism

“Concurrency is about *dealing with* lots of things at once. Parallelism is about *doing* lots of things at once.”
— Rob Pike, 2012

Concurrency

- Structure the program as independent tasks
- Many tasks are in flight, but they take turns
- You want responsiveness and overlapped I/O
- Works on one core: the OS hands out time slices

? QUESTION FOR YOU

One thread on one core downloads three files at once. Concurrency or parallelism?

Parallelism

- Run independent work at the same time
- One goal only: finish sooner (speedup)
- Needs real hardware — separate execution units
- Lives inside one core too: SIMD, superscalar issue

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→ ANSWER

Concurrency. The requests just sit on the network and the core waits on I/O — nothing is computed at the same time.

counter++ is really three instructions

One line in your source, three separate steps for the CPU.

? QUESTION FOR YOU

Two threads run counter++ a million times each. What number gets printed?

C++

```
int counter = 0;           // shared variable
void work() {
    for (int i = 0; i < 1000000; ++i)
        counter++;
}
std::thread a(work), b(work);
a.join(); b.join();
std::cout << counter << "\n"; // 2000000?
```

Java

```
static int counter = 0;
static void work() {
    for (int i = 0; i < 1000000; i++)
        counter++;
}
Thread a = new Thread(App::work), b = new Thread(App::work);
a.start(); b.start(); a.join(); b.join();
System.out.println(counter); // 2000000?
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```

One increment, as GCC expands it on x86-64

```
mov  eax, DWORD PTR counter[rip] ; 1. LOAD
add  eax, 1                      ; 2. ADD
mov  DWORD PTR counter[rip], eax ; 3. STORE
```

The thread can be descheduled between any two steps, and a second core can cut in right in the middle.

Java

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static int counter = 0;
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System.out.println(counter); // 2000000?
```

→ ANSWER

Anything from 2 to 2 000 000. On my laptop it printed **1 187 431** — the increments overwrite each other, and the number is different every run.

Where the lost increment goes

One bad interleaving and two increments add up to +1 instead of +2.

Step	Thread A	Thread B	counter in memory
1	load eax = 0	—	0
2	—	load eax = 0	0
3	add eax = 1	—	0
4	—	add eax = 1	0
5	store counter = 1	—	1
6	—	store counter = 1	1

Both increments ran, yet memory holds 1: step 6 blindly overwrote step 5.

! DON'T DO THIS

Never assume `counter++` is atomic. It is a *data race*: undefined behavior in C++, and in Java a silent stream of lost updates.

Task, data, pipeline: three shapes of parallel work

Almost everything you parallelize is one of these three shapes, or a mix of them.

1. Task parallelism

different jobs, different code

Task A (CPU 0)

Task B (CPU 1)

- Each thread runs different code
- They may share the same data
- Balance the jobs, or one thread lags

Example: a web server

2. Data parallelism

one operation, many data chunks

D_0 | D_1 | D_2 | D_3

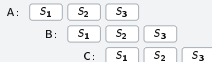
↓ the same $f(x)$

- Every thread takes its own chunk
- All of them run the same operation
- Equal chunks and cache locality set the speed

Example: adding two arrays

3. Pipeline parallelism

stages run at the same time



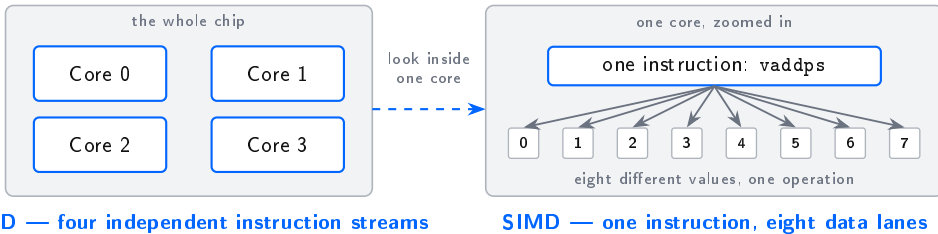
- One item walks through every stage
- Different items sit in different stages
- The slowest stage sets the pace

Example: a compiler

Real programs mix all three — and each shape invites its own thief: **task** loses to imbalance, **data** to the memory system, **pipeline** to its slowest stage.

One chip, two kinds of parallelism

Asked how much parallelism a chip offers, almost everyone answers with the core count. That is one axis of two.



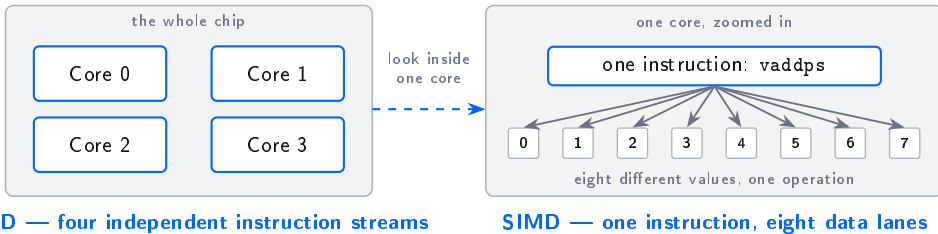
Flynn named four classes like this back in 1966. The fourth one, MISD, never really happened outside textbooks.

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Four cores, eight lanes each. How many float additions can this chip retire at once — and how many does your loop actually use?

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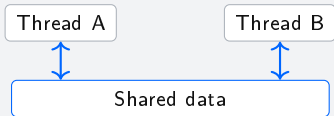
Thirty-two. A plain scalar loop on one thread uses **one** of them — and leaves 97% of the chip idle before a single lock is taken.

Shared memory or messages: how the pieces talk

The work is split. Now the parts have to exchange data — and you get to pick how.

Shared memory

everyone touches the same bytes



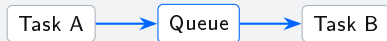
Threads read and write the same data.

Fast, but every concurrent write must be coordinated.

Example: a shared counter.

Message passing

nothing is shared, data is sent



Tasks send and receive messages.

The queue hands ownership over, so two tasks never share a payload.

Example: a job queue.

■ RULE OF THUMB

If handing off a task costs more than doing it, your tasks are too small — batch them.

Amdahl's law: the serial part sets the ceiling

First thief: the algorithm. Parse a file, then crunch the records independently.

One core: 100 seconds

5 s: parsing the file, record by record.

In this program that part is strictly serial.

95 s: crunching the records.

We split the records evenly across cores.

Start with two cores

Parsing: still 5 s. Crunching: $95/2$ s.

$$T(2) = 5 + 95/2 = 52.5 \text{ s}$$

$$\text{Speedup: } 100/52.5 \approx 1.90\times.$$

$$\text{For } N \text{ cores: } T(N) = 5 + \frac{95}{N} \text{ s, } S(N) = \frac{T(1)}{T(N)} = \frac{100}{5 + 95/N}.$$

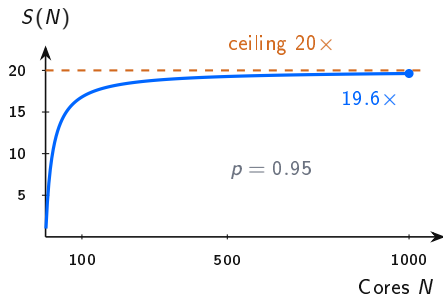
Same file every time, and we assume a perfect split with zero overhead.

? QUESTION FOR YOU

How much faster does this job run on 1000 cores? Could it ever finish in under 5 seconds?

Amdahl's law: the serial part sets the ceiling

First thief: the algorithm. Parse a file, then crunch the records independently.



→ ANSWER

$$T(1000) = 5 + 95/1000 = 5.095 \text{ s.}$$

$$S(1000) = 100/5.095 \approx 19.6\times.$$

You can never beat 5 s: the parsing never goes away.

The general form: Amdahl's law

$p = 0.95$: the share of one-core time that is parallel.

$$T(N) = T(1)((1 - p) + p/N)$$

$$S(N) = \frac{1}{(1 - p) + p/N} \xrightarrow{N \rightarrow \infty} \frac{1}{1 - p} = 20$$

? QUESTION FOR YOU

How much faster does this job run on 1000 cores? Could it ever finish in under 5 seconds?

Gustafson: what if the problem grows with the cores?

Second scaling law: hold the clock fixed and grow the problem instead.

Amdahl = strong scaling

The problem size is fixed and you want it sooner: a render, a build.

You can never pass the ceiling $1/(1 - p)$.

Gustafson = weak scaling

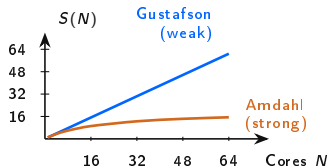
The time budget is fixed and you solve a bigger problem: a finer mesh, 4K instead of HD.

$$S(N) = (1 - p) + p \cdot N$$

With $p = 0.95$: $S(64) = 60.9\times$.

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Ask first: am I holding the problem fixed, or the clock?



? QUESTION FOR YOU

Cores doubled, clock unmoved, twice the data crunched. Is that a win?

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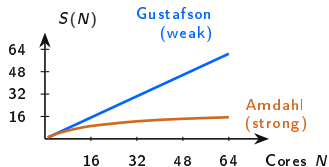
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→ ANSWER

Yes — weak scaling. The serial part stayed fixed; only the parallel work grew.

Karp–Flatt: measure your own serial fraction

Amdahl needs p . Here is how you measure it on a program you already have.

? QUESTION FOR YOU

You measured $4.2\times$ on 8 threads. Good result or bad result?

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The serial fraction you measure:

$$e = \frac{1/S(N) - 1/N}{1 - 1/N}$$

All you need is measured $S(N)$; $e \approx (1 - p)$.

■ RULE OF THUMB

One e tells you nothing. Plot e as N grows.

e flat $\Rightarrow$ serial code is the wall. Go parallelize it.

e grows $\Rightarrow$ overhead is piling up (the next three thieves).

Your measurements, and e from them:

N	$S(N)$	e
1	1.00	—
2	1.89	0.058
4	3.25	0.077 $\uparrow$
8	4.20	0.129 $\uparrow$
16	5.10	0.142 $\uparrow$

At $N = 8$: $e = (1/4.2 - 1/8)/(1 - 1/8) = 0.129$.

→ ANSWER

The $4.2\times$ alone tells you nothing. e climbs 0.058 $\rightarrow$ 0.142: overhead is growing. Profile before blaming the locks.

What happens after the plateau

Amdahl promises a plateau. Real machines do something else.

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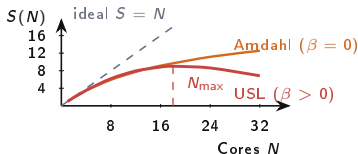
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$$S(N) = \frac{N}{1 + \alpha(N - 1) + \beta N(N - 1)}$$

α is waiting in queues, β is coherence chatter. Set $\beta = 0$ and you are back to Amdahl.

How to get them: measure $S(N)$ up to 32 cores and fit by least squares. Quick check: a peak at $N_{\max}$ means $\beta \approx 1/N_{\max}^2$.

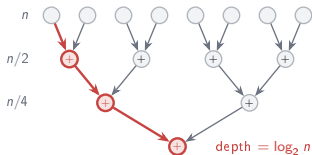
High $\alpha \Rightarrow$ something is serialized: shorten it, or split the shared resource. High $\beta \Rightarrow$ stop sharing mutable state.

→ ANSWER

It can. The penalty $\beta N(N - 1)$ grows quadratically and drags the whole curve down.

Work and span: the algorithm's own ceiling

Work is every operation; span is the longest chain that waits on itself. Cores remove work, never the chain.



$T_1 = n$ (work), $T_\infty = \log_2 n$ (span, the red path). Parallelism $= n / \log_2 n$.
At $n = 10^6$: $T_\infty \approx 20$, so parallelism $\approx 50\,000$ — the point where extra cores stop helping, not a speedup. On 8 cores the algorithm is not what holds you back.

Two quantities, two floors:

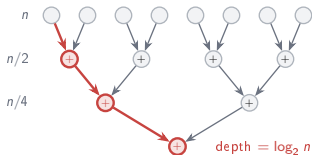
T_1 (work) — every operation, counted on one core. T_∞ (span) — the dependency chain you still walk on infinitely many cores.
Work law: $T_N \geq T_1/N$. Span law: $T_N \geq T_\infty$ — no hardware breaks that floor.

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So what is the span of a plain loop that does `sum += a[i]`?

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→ ANSWER

$T_\infty = n$: every step waits for the one before it, so the parallelism is exactly 1.

Roofline: are you limited by math or by memory?

Two ceilings: how fast you can compute, and how fast memory can feed you. The work you do per byte decides which one you hit.

Limit = $\min(\text{peak FP}, \text{BW} \times I)$, where I is FLOPs per byte moved from DRAM.

Vector add: $I = 1/12 \approx 0.08$

```
for (int i = 0; i < n; ++i) a[i] = b[i] + c[i];  
// 1 FLOP; bytes: b 4 + c 4 + a 4 = 12
```

Hopelessly memory-bound: the cores sit idle waiting for data.

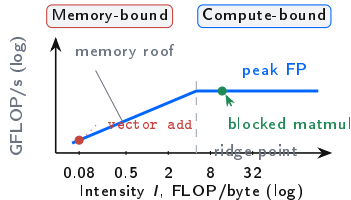
Blocked matrices, $B \times B$: I grows with B

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// one B x B block: "2*B^3 FLOP per "3*B^2*4 bytes  
// I ~ B/6: B=8 -> 1.3, B=64 -> 10.7
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Same algorithm, different traversal order — and now it is compute-bound.

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Find the ridge point first. Left of it, raise I ; right of it, chase FLOPs.



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You sit left of the ridge point. Will doubling your threads help?

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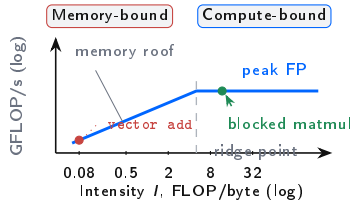
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→ ANSWER

No — the memory bus is the wall. Raise I first, then add threads.

Memory hierarchy: what a cache miss really costs

Second thief: the hardware. It takes nothing out of your algorithm — it just makes it wait.

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A core needs one number that is not in any cache. How long does it wait?

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→ ANSWER

Roughly **200 cycles**. Independent work keeps flowing — but everything that **depends on that value** stops dead until it lands.

Measured latencies (7-cpu.com), 3 GHz

Level	Skylake	Zen3	Total, cycles
L1d (32 KB)	4 cycles	4 cycles	≈ 4
L2 (256 / 512 KB)	12 cycles	12 cycles	≈ 12
L3 (8 / 16 MB)	42 cycles	47 cycles	≈ 45
RAM (DDR4)	42 + 51 ns	47 + 74 ns	190 / 270

Last row = the full cost of a miss: DRAM plus the 42 cycles already spent missing in L1–L3. So “about 200” is 190 on this i7-6700, 270 on this Ryzen 5600G.

What this means in code

Walk row by row and the whole line pays off: 64 B is 8 numbers at once. Walk column by column and every element costs a new line and a new miss — same algorithm, several times the runtime.

One caveat: the famous “Latency numbers every programmer should know” (L1 3 cycles, mutex 50, RAM 100 ns, cross-datacenter 500 μs) is a teaching aid. Learn the ratios from it, never the absolute numbers.

Cache lines and MESI: reads are cheap, writes are not

A core cannot own one byte. The smallest thing the hardware moves is a whole 64-byte line.

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Two cores read the same line happily. One now **stores** to it — what happens to the other copy?

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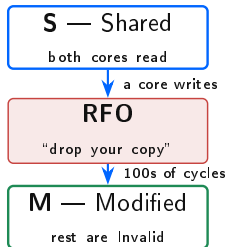
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→ ANSWER

It dies. Reads are free: both copies sit in **S (Shared)**. The write costs an **RFO** — claim the line, **invalidate every other copy**, then modify.



Why one byte costs a whole line

The core must own the entire 64-byte line before it touches one byte: broadcast, wait for every copy to drop, then write.

Two consequences, and both bite

- False sharing (next slide) — you pay in full even for different variables;
- Atomics and locks (Act IV) — a shared counter on 16 cores is slower than on one.

MOESI (AMD64) adds O (Owned): a dirty line goes core-to-core, skipping DRAM.

False sharing: one cache line, two victims

The worst bug in this act: the code looks perfectly parallel and runs like it is serial.

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Two threads write to **different** variables. Nothing is shared, no locks anywhere. Can they still cripple each other?

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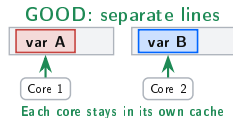
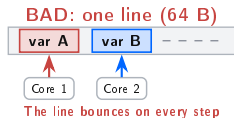
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→ ANSWER

They can — if both land in the **same 64-byte line**. The hardware sees lines, not variables, so every write rips the line away from the other core.



How far it drops

- **Drepper (4×P4):** +390/734/1147 % on 2/3/4 threads — worse with every thread.
- Adding threads makes it slower, not faster.

How to fix it and how to find it

- C++: `alignas(64)`. Java: `@Contended` (JDK 8).
 - Find it: `perf c2c` names the hot line.
- Trap:** hand-made padding gets optimized away.

Reordering: why “0 0” is a legal output

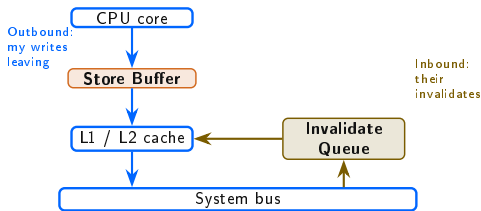
Your CPU does not run your program in order, and neither does your compiler.

Initially $x = y = 0$

```
// thread 1      // thread 2
x = 1;           y = 1;
print(y);        print(x);
```

? QUESTION FOR YOU

Try to get “0 0” by interleaving these four lines. Whichever order you pick, one store always lands first — so is “0 0” impossible?



- Your write waits in the store buffer, so your neighbor still sees the old value.
- Their invalidate waits in the invalidate queue, so you still read your stale copy.
- A fence does not “flush” these queues — it *constrains the order*: nothing after it may become visible before what came ahead of it.
- This is the theft: reordering is how the hardware buys speed, and every fence you add hands part of it back.

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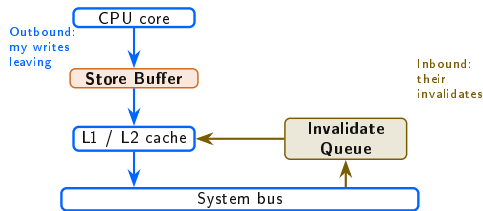
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It happens on real x86 (**StoreLoad reordering**). Each core parks its own store in a store buffer and races ahead to the load — so both loads can run **before either store is visible** to the other core.



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Eight cores, and still not eight times

No locks, no sharing, perfectly independent threads — and the curve still flattens.

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Eight threads each stream through their own huge array. At how many threads does the speedup stop growing — and what decides that number?

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Nothing. Saturated channels mean extra threads just queue for the same bytes. Measure **where** your curve flattens.

1. Bandwidth — the real ceiling

All cores share a **fixed number of channels**. Saturate them and more threads add queuing, not throughput. Measure with **STREAM** (McCalpin) and find the knee of your curve.

2. SMT hides stalls only

+5 % of die area buys +15–30 % when one thread stalls and the other has work. But both share the ALUs and L1: if the core waits for **memory**, the second waits too — SMT adds no bandwidth.

In HPC it often loses. Measure both

3. NUMA splits the road in two

A multi-socket server has more than one set of channels, and the **other** socket's memory is slower:

```
numactl --hardware  
numactl --cpunodebind=0 ./app
```

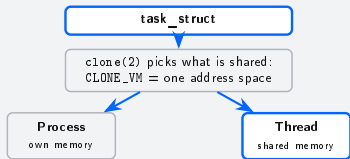
■ RULE OF THUMB

The hardware taxes you **per line** and **per byte**. Next thief: the OS, which taxes you **per thread**.

On Linux, a thread and a process are the same object

Third thief: the OS. Both are one `task_struct`; `clone(2)` decides what they share.

One structure, two modes



What to take away

- Threads share memory **by default** — that is where races come from.
- Processes are isolated, but pay for **expensive IPC**.
- A process is really a **thread group**; every thread in it costs real kernel resources.

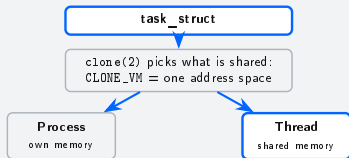
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Why does Chrome isolate sites in separate **processes** instead of one shared address space?

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? QUESTION FOR YOU

Why does Chrome isolate sites in separate **processes** instead of one shared address space?

→ ANSWER

Isolation. A crash in one site, or an exploit inside it, must not reach the rest of the browser. Chrome pays for that with expensive IPC, and it pays on purpose.

A million threads: why the OS says no

? QUESTION FOR YOU

`ulimit -s` gives every thread an 8 MB stack. How many threads fit into 16 GB of RAM?

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→ ANSWER

Not 2000 — far more. Those 8 MB are only a virtual reservation; pages are handed out when you actually touch them. You hit the kernel and the scheduler long before you run out of RAM.

Virtual is not physical

- 8 MB (`ulimit -s`) is just an **address range**, not memory you own.
- Physical memory (**RSS**) arrives page by page — only where you really wrote. Usually tens of KB per thread.

Where the real wall is

- Context switches, L1-L3 misses and TLB flushes.
- The scheduler: run queues, load balancing, migrations between cores.
- Hard OS limits: `RLIMIT_NPROC`, `threads-max`, `pid_max`.

What a thread costs: pthread, goroutine, Erlang, virtual

The lighter the thread, the more of them a runtime can carry — and the less work the OS kernel has to do.

pthread (OS)

Stack: 8 MB reserved by default; you can lower it only to `PTHREAD_STACK_MIN = 16 KB`.

A kernel object; the kernel picks who runs next. In practice: thousands.

Goroutine (Go)

Stack: starts at 2 KB and grows or shrinks on demand.

Scheduled in user space. Hundreds of thousands per process.

Erlang process

Stack+heap: 338 words at birth, about 2.7 KB.

Each one owns its heap. Default ceiling: 1 048 576.

Java virtual thread

The stack lives on the heap as a stack chunk, sized as needed.

Millions of tasks; the JVM schedules them in user space.

■ RULE OF THUMB

Pick the thread that matches the job: kernel threads for cores, user-space threads for waiting.

What virtual threads actually buy you

A million of them fit where a few thousand platform threads did. The interesting question is what that gets you.

When to reach for them

- **Yes:** thousands of tasks that spend their life **waiting on I/O**.
- **No:** CPU-bound work — you still stop at the number of physical cores.
- 10 000 tasks doing “sleep 1s”: a pool of 200 platform threads finishes ~200 per second, virtual threads ~10 000.

Traps to watch for

- **Pinning:** on JDK 21–23, blocking inside synchronized nailed the virtual thread to its carrier. JEP 491 fixed that in **JDK 24**.
- “*Virtual threads should never be pooled*” — pool the scarce resource, never the thread.
- A `ThreadLocal` holding something heavy will flood the heap.

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Your workload is pure number crunching on 8 cores. You move it to a million virtual threads. What changes?

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→ ANSWER

No. They scale waiting, not computing. Nothing about them adds cores to your machine.

The scheduler: who takes the CPU away from whom

A cooperative scheduler waits for the task to be polite. A preemptive one just takes the CPU back.

Go: the price of being polite

Before Go 1.14 a goroutine yielded only at a function call. So **a tight loop with no calls froze the whole runtime**.

Go 1.14 (2020) made preemption arrive as an OS signal: *“loops without function calls no longer deadlock the scheduler”*.

Linux always preempts; only the policy changed — EEVDF replaced CFS in kernel 6.6.

What a context switch costs

Case	Latency
OS thread, same core	1.2–1.5 μ s
OS thread, moved to another core	$\sim$ 2.2 μ s
Goroutine (user space)	$\sim$ 170 ns

The real bill is hidden: L1–L3 and the TLB go cold, and re-warming them costs more than the switch.

■ RULE OF THUMB

Oversubscribe — far more threads than cores — and the CPU burns its time shuffling them. Start from one rule: a CPU-bound pool $\approx$ the core count.

Priority inversion: guess who wins

Three threads with three different priorities, and exactly one shared mutex between them.

? QUESTION FOR YOU

The low-priority thread holds the mutex, the high-priority one waits for it. A medium thread that never touches the mutex is also runnable. Who gets the CPU?

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→ ANSWER

The medium one — the thread that has nothing to do with the lock. As long as it stays runnable it keeps preempting the lock holder, so the lock is never released and the high-priority thread misses its deadline.

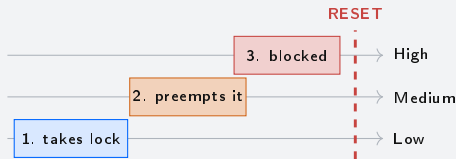
This hits your code, not just spacecraft

- **Thread pools:** a background task holds the lock a web request needs, while every worker is busy elsewhere. GC and service threads steal CPU from the lock holder too.
- **nice:** lower the priority of whoever owns a shared resource and you have just stalled the entire service.
- **The cure:** priority inheritance (PTHREAD_PRIO_INHERIT), fewer shared locks, shorter critical sections.

The same bug on Mars: Pathfinder, 1997

The lander touched down, started driving — and then began rebooting itself, over and over, from exactly the scenario we just walked through.

The 1553 bus, 8 Hz cycle



Low task locks the bus → medium task preempts it → high task blocks → **the watchdog resets the lander.**

Diagnosis, patch, lesson

- Logging saved them: a ring trace buffer let the lab on Earth reproduce the bug in 18 hours.
- The fix was uploaded to Mars: turn **priority inheritance** back on in `selectLib()`. Wind River had shipped it off *for speed*.
- “When you fly COTS stuff — make sure you know how it works” (G. Reeves).